

Forecasting the S&P SL 20 Index

ARIMA vs. LSTM vs. Hybrid — One-Step and Multi-Step Analysis

A complete record of two forecasting experiments on the Colombo Stock Exchange S&P SL 20 index (2015–2026), benchmarked against the source paper of Vithushan & Kethmi (IRC-OUSL 2025).

Index	S&P SL 20 (Colombo Stock Exchange)
Data window	2015-01-01 to 2026-04-16 (2,664 trading days)
Test window	2024-01-22 to 2026-04-16 (533 trading days)
Models	ARIMA, LSTM, Hybrid ARIMA+LSTM, Naive baseline
Experiments	1 = one-step-ahead 2 = multi-step (paper-style)
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1. Project Overview

This project forecasts the daily **S&P SL 20** index of the Colombo Stock Exchange (CSE) and compares three modelling setups on **identical data and an identical train/test split**:

- **ARIMA** — a traditional linear time-series model.
- **LSTM** — a recurrent neural network for nonlinear patterns.
- **Hybrid ARIMA + LSTM** — ARIMA models the linear part; an LSTM models ARIMA's residual (nonlinear leftover); the two are summed.

The work both **reproduces** and **extends** the source paper. We reproduce its ARIMA-vs-LSTM comparison, but on a newer dataset that **includes the 2020–2023 Sri Lankan crisis years** (which the paper deliberately excluded), and we add the **Hybrid** setup the paper described but never actually reported.

Crucially, we run the comparison **two ways**, because how you forecast the test window changes the numbers by an order of magnitude:

- **Experiment 1 — one-step-ahead ('easy mode')**. To predict each day, the model is given the *true* recent history up to the day before.
- **Experiment 2 — multi-step ('hard mode', the paper's way)**. The model forecasts the whole test window forward using *only its own predictions* — it never sees a real test value. This is what makes our numbers comparable to the paper's Table 4.

A **naive persistence baseline** is included throughout so we can tell whether the models add genuine skill or merely echo yesterday's price.

2. Data and Train/Test Split

All modelling uses **spsl20_trading_days_clean.csv** — the univariate S&P SL 20 closing level with weekends and carry-forward holiday duplicates removed (those fake flat days mislead both models). The split is **80/20 chronological** (never shuffled — this is sequential market data), computed once and reused by every script so all models train and are graded on exactly the same dates.

Portion	Rows	Share	Dates
Training	2,131	80%	2015-01-01 to 2024-01-19
Test (held out)	533	20%	2024-01-22 to 2026-04-16

Where the crisis sits: the volatile 2020–2023 crisis years fall inside the *training* window. The models are therefore trained through the turbulence and graded on the calmer, strongly rising 2024–2026 recovery — the deliberate stress test of this project. **No leakage:** every MinMax scaler is fit on the training portion only, and ARIMA parameters are estimated on training data only.

3. The Source Paper — Findings

Vithushan, M. & Kethmi, G. A. P. — *A Comparative Analysis of Time Series Models for Predicting the S&P SL 20 Index of the CSE*, IRC-OUSL 2025.

What the paper did:

- Used daily S&P SL 20 data for **2010–2018** (~2,160 observations), **deliberately excluding** later years because of ‘unusual circumstances’ (the crisis).
- Compared **ARIMA vs. LSTM** only — a two-way study. A hybrid is mentioned in the introduction but **never reported** in the results.
- ARIMA: log-transformed series, ADF $p = 0.05108$ (non-stationary), order **(2,1,0)** chosen from ACF/PACF. LSTM: Adam optimiser, MSE loss.
- Split by date: train 2010–2016, test 2017–2018.

The paper’s results (Table 4):

Model	MAE	MAPE	RMSE
ARIMA	233.96	7.04%	269.57
LSTM	249.37	6.96%	269.86
Hybrid	not reported	—	—

The paper’s conclusion: ARIMA and LSTM perform similarly. ARIMA is marginally better on MAE and RMSE (suited to linear trends); LSTM is marginally better on MAPE (captures nonlinear patterns). The paper notes ARIMA produced ‘constant predictions in some periods’ — a fingerprint of the flat multi-step forecast that becomes important in our Experiment 2.

4. Experiment 1 — One-Step-Ahead Forecast

Method. Each test-day prediction uses the true prior history. For the LSTM/Hybrid, the previous w real days (or residuals) feed the next-day prediction. For ARIMA, parameters are frozen after training and the model *state* is updated with actual test observations (`statsmodels.append(refit=False)`) — this is **not** retraining and adds no parameter leakage. Model details:

- **ARIMA:** ADF-based $d = 1$; `auto_arma` selected **ARIMA(1,1,4)** (lower AIC than the (2,1,0) baseline). Residual diagnostics: Ljung-Box $p = 0.998$ (no autocorrelation), Breusch-Pagan $p = 0.106$ (homoskedastic), Jarque-Bera $p < 0.0001$ (non-normal — one extreme crisis-era residual spike).
- **LSTM (PyTorch):** 2 stacked LSTM layers $\times$ 50 units, dropout 0.2, `Dense(1)`, MSE loss, Adam, early stopping on a validation slice; MinMax scaler fit on train only; seeds fixed. Window 60 selected.
- **Hybrid:** ARIMA(1,1,4) forecast + an LSTM trained on ARIMA’s training residuals (window 30).

Experiment 1 results (test window, 533 days; lower is better):

Model	MAE	MAPE	RMSE
Naive (one-step)	37.06	0.788%	56.16

Model	MAE	MAPE	RMSE
ARIMA (1,1,4)	36.42	0.770%	56.25
LSTM (window 60)	140.52	2.491%	198.10
Hybrid (best)	36.25	0.766%	56.15

Reading it: the Hybrid wins on all three metrics, but only marginally ahead of pure ARIMA — and both barely beat the naive baseline. At a one-day horizon the index is almost a random walk (tomorrow $\approx$ today), so persistence alone already scores $<1\%$ MAPE. The standalone LSTM lags the price and is the clear laggard. The Hybrid adds almost nothing over ARIMA because ARIMA's residuals are essentially white noise (Ljung-Box $p = 0.998$), leaving no structure for the residual-LSTM to exploit.

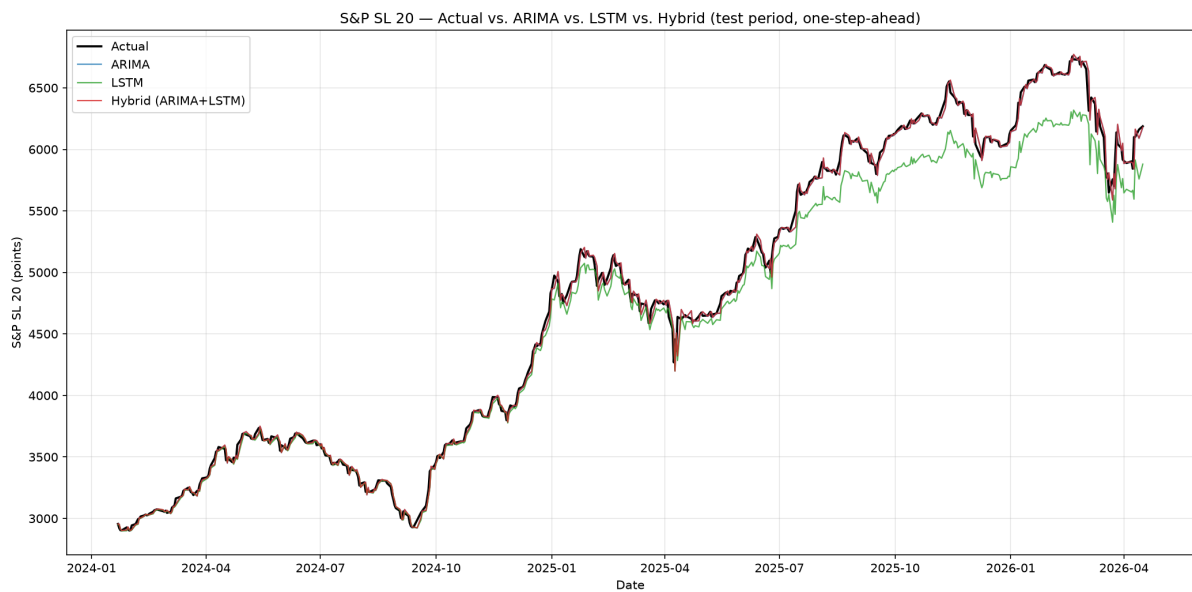


Figure 1. Experiment 1 (one-step). ARIMA and Hybrid sit on the actual line (ARIMA hidden under Hybrid); the standalone LSTM lags below during the uptrend.

5. Experiment 2 — Multi-Step (Paper-Style) Forecast

Method. The model is cut off from real test values and must roll the whole 533-day horizon forward on its own guesses — the harder, honest test the paper used.

- **ARIMA:** paper recipe — log-transform, $d = 1$, baseline (2,1,0); auto_arima selected **ARIMA(0,1,1)**. A single dynamic forecast for the entire horizon (`get_forecast(steps=533)`); no test value fed back; log inverted before scoring.
- **LSTM (recursive):** seeded with the last 60 training days, predicts day 1, then feeds each prediction back as the newest input and rolls forward — the window fills with the model's own outputs.
- **Hybrid:** multi-step ARIMA forecast + a residual-LSTM rolled forward recursively.
- **Naive (flat):** every test day = the last training value — the honest 'no-skill' reference for multi-step.

Experiment 2 results (test window, 533 days; lower is better):

Model	MAE	MAPE	RMSE
Naive (flat)	1788.9	33.23%	2162.5
ARIMA (0,1,1) log	1792.8	33.32%	2165.9
LSTM (recursive)	1913.9	35.93%	2277.0
Hybrid (multi-step)	1792.1	33.30%	2165.3

Reading it: every multi-step forecast collapses to a near-flat line around 3000 while the index climbs to ~6700. The **naive flat line is actually the best** (lowest RMSE); ARIMA and the Hybrid essentially tie it, and the recursive LSTM does slightly worse. **None of the models adds real multi-step skill** over 533 days — in a trending market, forecasting the whole horizon from a single origin is close to hopeless, and 'flat at the last value' is a hard reference to beat.

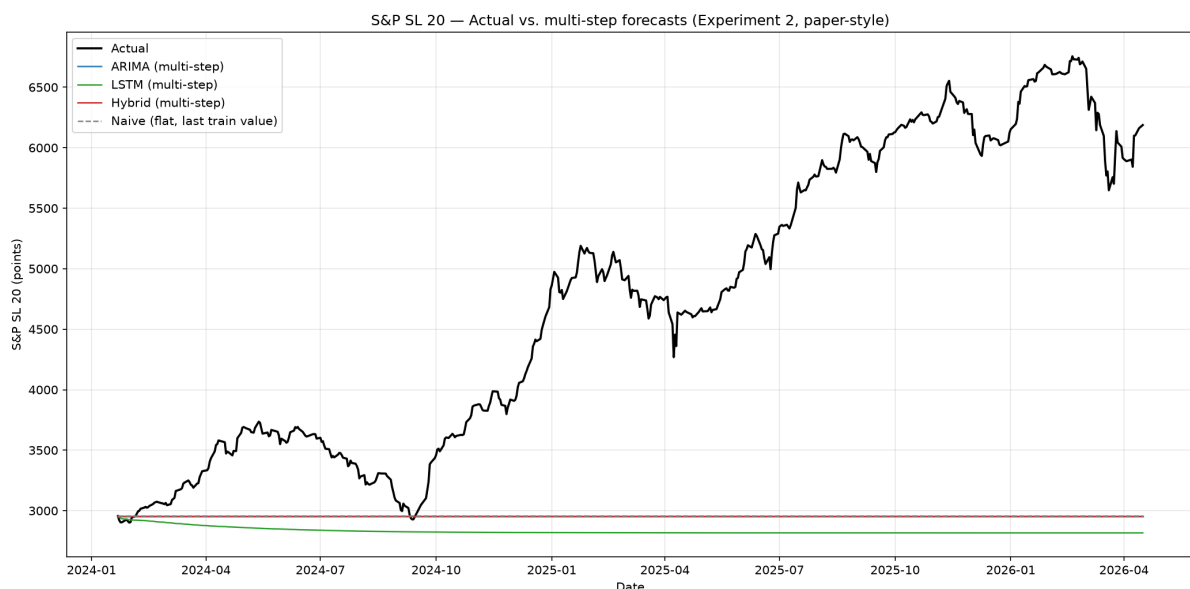


Figure 2. Experiment 2 (multi-step). All forecasts flatten near 3000 (ARIMA, Hybrid and Naive overlap; the LSTM drifts slightly lower) while the actual index rises to ~6700.

6. Combined Comparison — One-Step vs. Multi-Step

The single most important table in this report. The same models, same data, same test window — only the *forecasting protocol* differs between the two blocks.

Model	1-step MAE	1-step MAPE	1-step RMSE	Multi MAE	Multi MAPE	Multi RMSE
Naive	37.06	0.79%	56.16	1788.9	33.23%	2162.5
ARIMA	36.42	0.77%	56.25	1792.8	33.32%	2165.9
LSTM	140.52	2.49%	198.10	1913.9	35.93%	2277.0
Hybrid	36.25	0.77%	56.15	1792.1	33.30%	2165.3

The ~40× jump in every error metric from left block to right block is the difference between the easy and the hard test — not a change in the models. This is exactly why one-step numbers must never be quoted as if they measured genuine forecasting skill.

7. How Our Numbers Compare to the Paper

Our **multi-step** block is the fair comparison to the paper's Table 4, because it uses the same forecasting method. Yet our errors are far larger (ARIMA MAPE 33% vs. the paper's 7%). **The cause is the test window, not the models.**

- The paper's 2017–2018 test period was roughly flat, so its multi-step forecasts — which collapse toward a flat line — stayed close to the truth.
- Our 2024–2026 window is a strong post-crisis bull run (~3050 → ~6700), so any flat or converging multi-step forecast is badly wrong. Same method, much harder horizon.
- Both studies agree on the qualitative headline: **ARIMA ≥ LSTM**. Our one-step result sharpens this (ARIMA clearly beats the standalone LSTM); the paper found them near-tied.
- We supply the evidence the paper lacked: the Hybrid *is* marginally best one-step, but its gain over ARIMA is negligible — ARIMA already whitens the residuals.

8. Conclusions and Key Takeaways

- **One-step:** Hybrid < ARIMA < Naive << LSTM. The Hybrid is best but the margin over ARIMA and even the naive baseline is tiny — daily persistence dominates.
- **Multi-step:** Naive ≤ Hybrid ≈ ARIMA < LSTM. No model beats the flat baseline; long-horizon forecasting of this trending index is effectively no-skill.
- **Beat the baseline?** Only marginally, and only one-step. The honest reading is that neither ARIMA, LSTM, nor the Hybrid delivers reliable multi-step skill on this series.
- **Crisis effect:** the 2020–2023 crisis (in the training window) taught a volatile, largely trendless regime, so the fitted ARIMA carries almost no drift — which is exactly why its multi-step forecast flattens and undershoots the rising recovery. It also left fat-tailed residuals (why RMSE » MAE throughout).
- **Honesty note:** sub-1% one-step MAPE looks impressive but mostly reflects the easy test; the multi-step block is the one to trust for real forecasting ability.

9. Environment and Reproducibility

Modelling ran in a local virtual environment under **Python 3.14** with pandas, numpy, statsmodels, pmdarima, scikit-learn, matplotlib, and **PyTorch**. **Note:** the project spec named TensorFlow/Keras for the LSTM, but Python 3.14 currently has no TensorFlow wheel, so the LSTM was implemented in PyTorch with the *identical* architecture and training recipe (2×50-unit LSTM, dropout 0.2, Dense(1), MSE, Adam, early stopping, fixed seeds). This is a library substitution only; the method is unchanged.

Script	Produces
01_arima.py / 02_lstm.py / 03_hybrid.py	Experiment 1 (one-step) predictions + metrics
04_compare_results.py	Experiment 1 table + overlay (Figure 1)
05_arima_multistep.py / 06 / 07	Experiment 2 (multi-step) predictions + metrics
08_compare_multistep.py	Naive baseline, combined table, overlay (Figure 2)
make_report.py	This PDF report

Build order 01 → 08. Every script reuses `split_info.json` so the train/test split is identical across all experiments.

Experiment 1 and Experiment 2 write to separate filenames and do not overwrite each other.